

Explainable artificial intelligence framework for cycle time prediction and an empirical study for smart semiconductor manufacturing

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Acknowledgements

This research is supported by the National Science and Technology Council, Taiwan (NSTC 113-2221-E-007-114-MY3; NSTC 114-2222-E-006-010; NSTC 114-2223-E-007-014) and Powerchip Semiconductor Manufacturing Corporation (PSMC), Taiwan.

Declaration of Interest statement

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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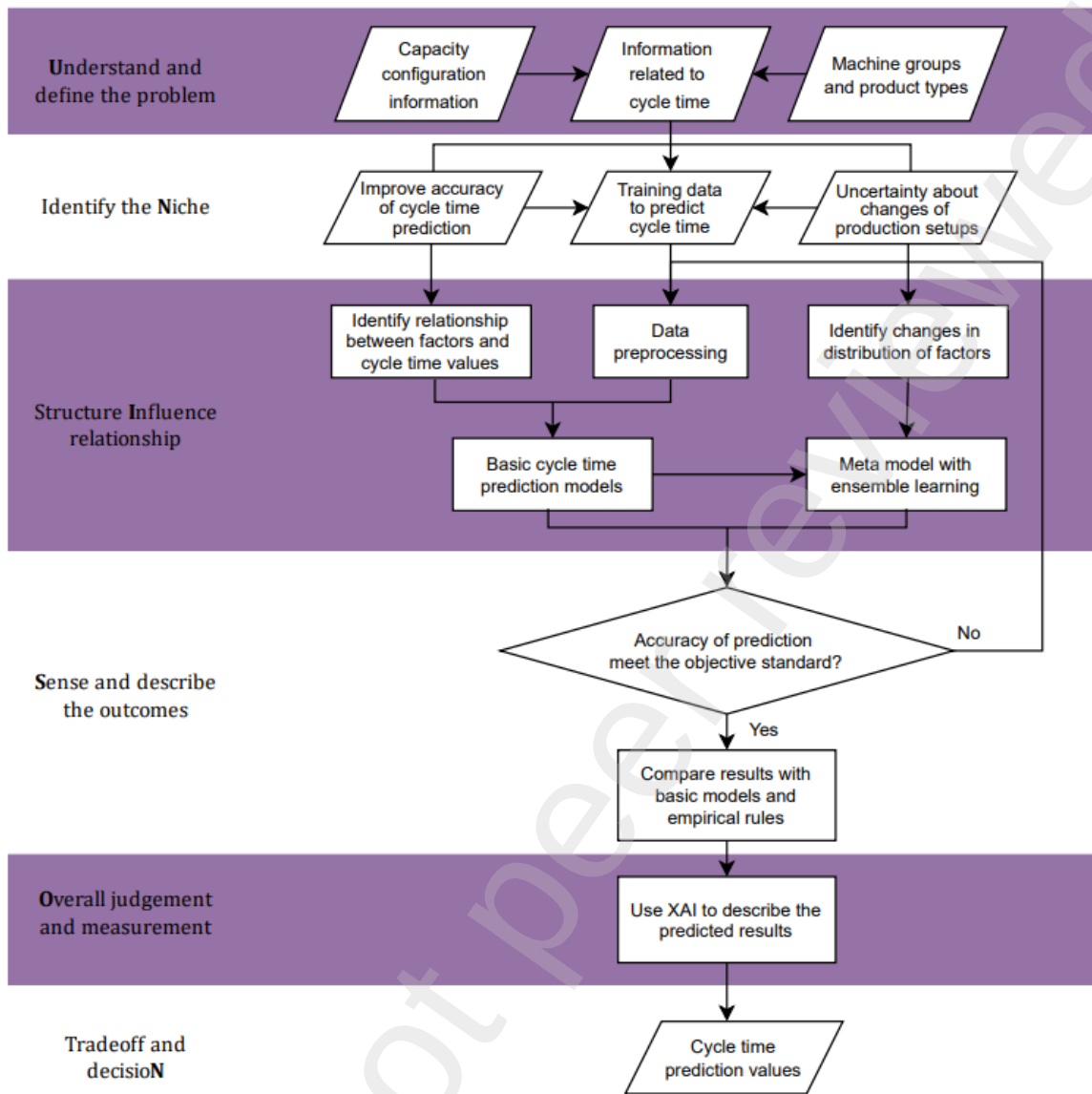


Figure 4 UNISON framework for explainable artificial intelligence based advanced cycle time prediction

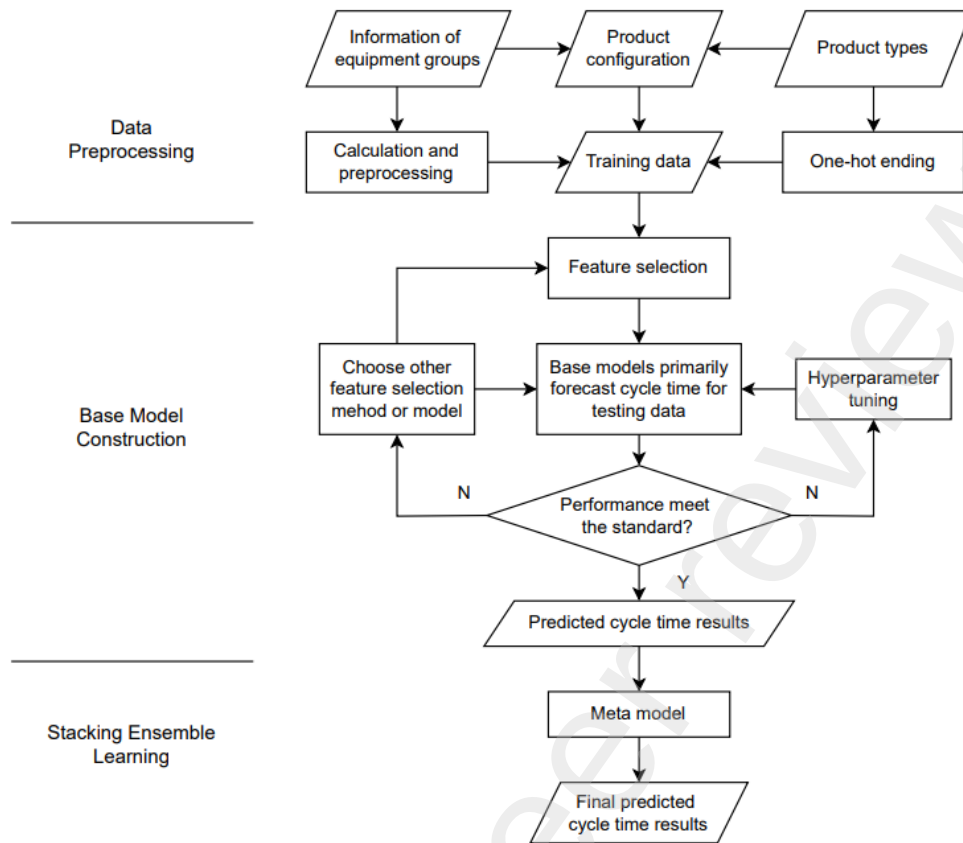


Figure 5 Three-stage framework for advanced cycle time prediction

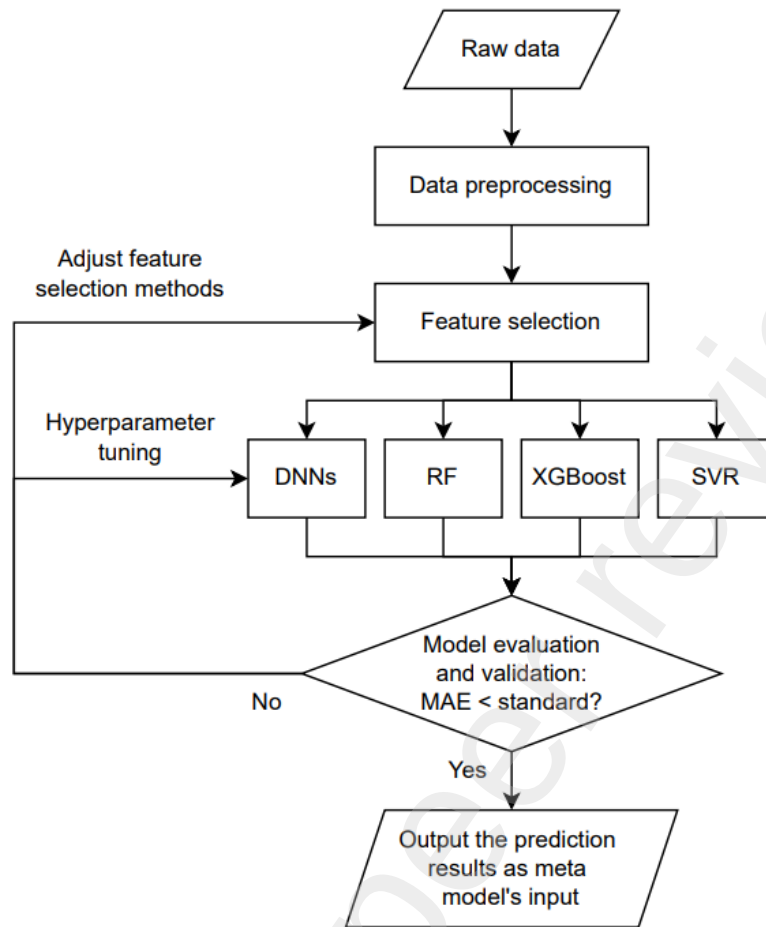


Figure 6 The architecture of base models' construction

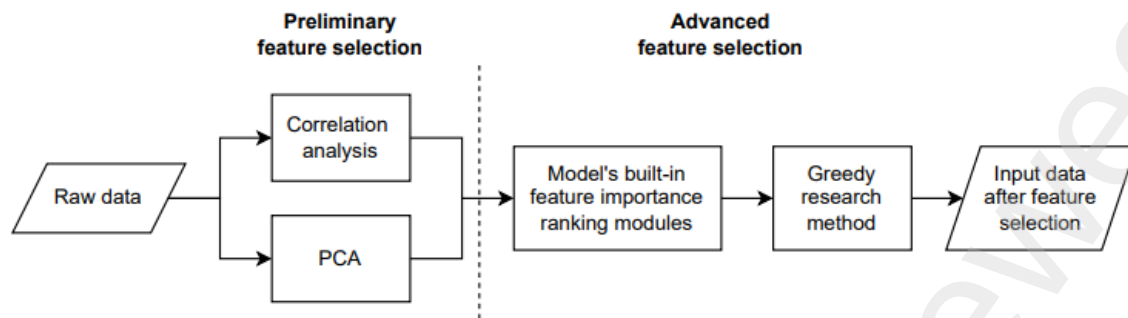


Figure 7 The flow chart of feature selection

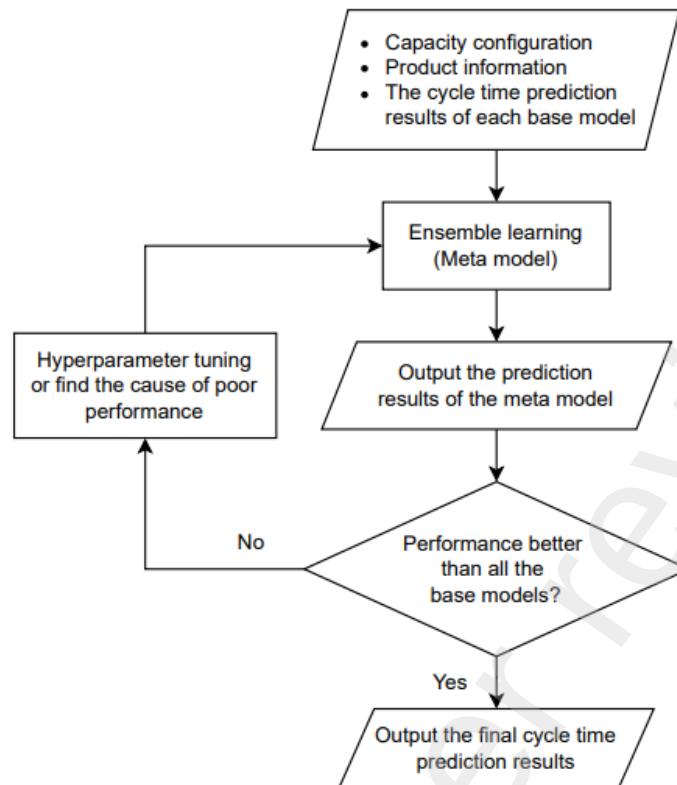


Figure 8 The architecture of meta model's construction

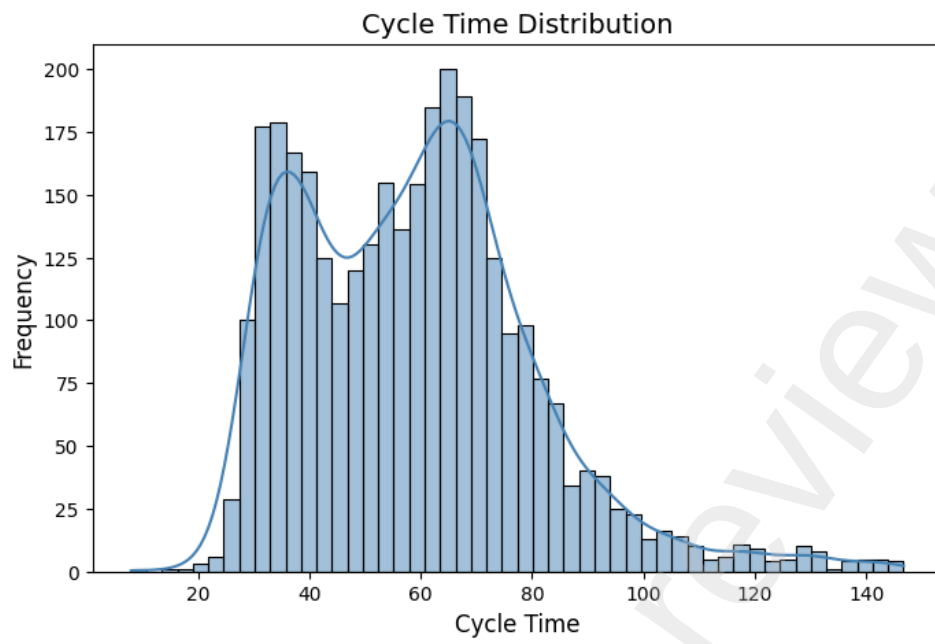


Figure 9 Distribution plot of cycle time

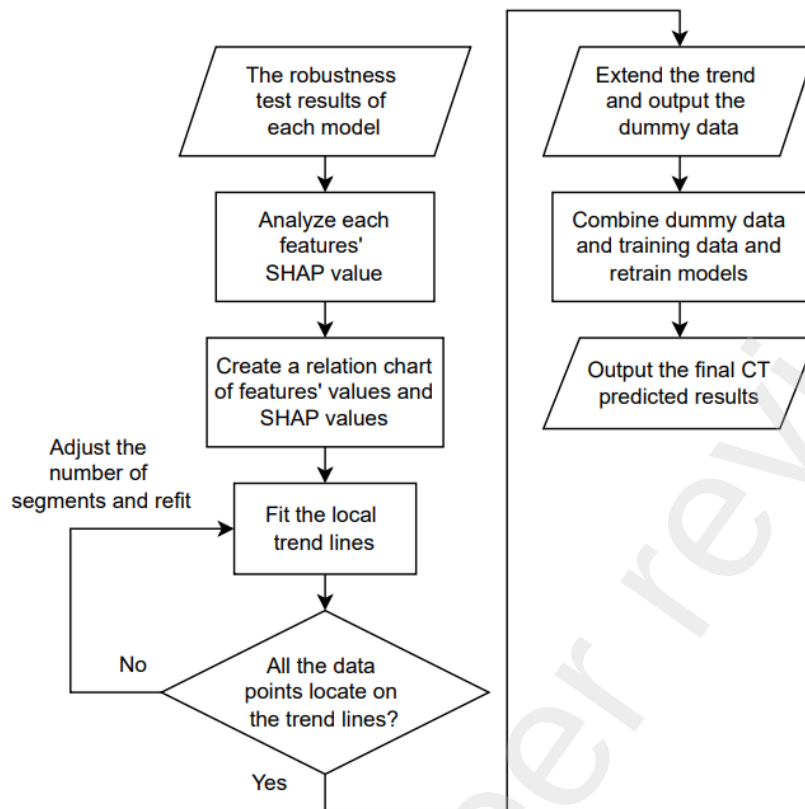


Figure 10 The flow chart of XAI model and dummy data construction

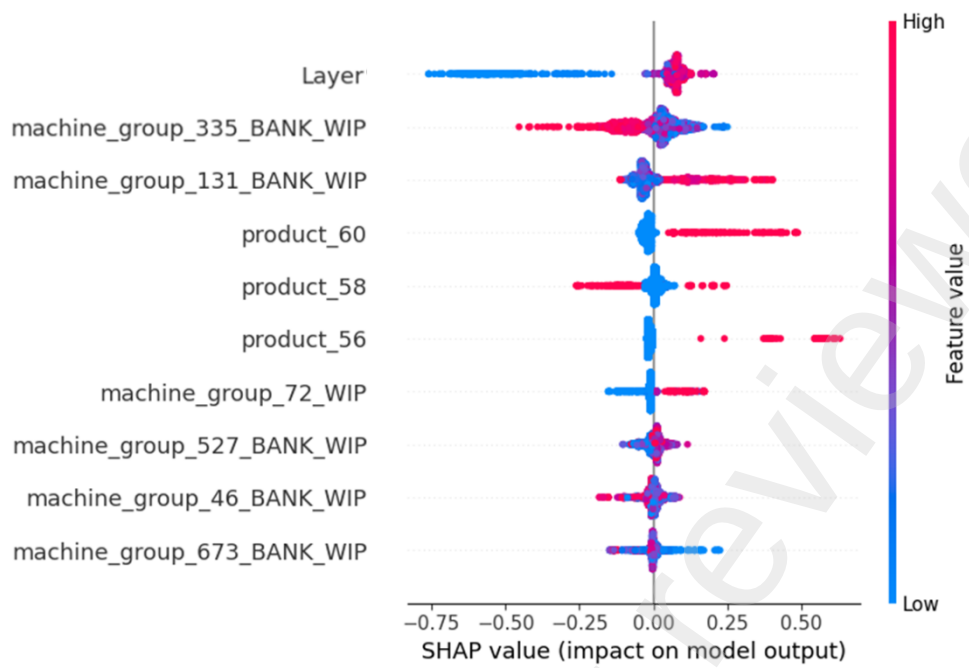


Figure 11 The summary plot of XAI model for Random Forest

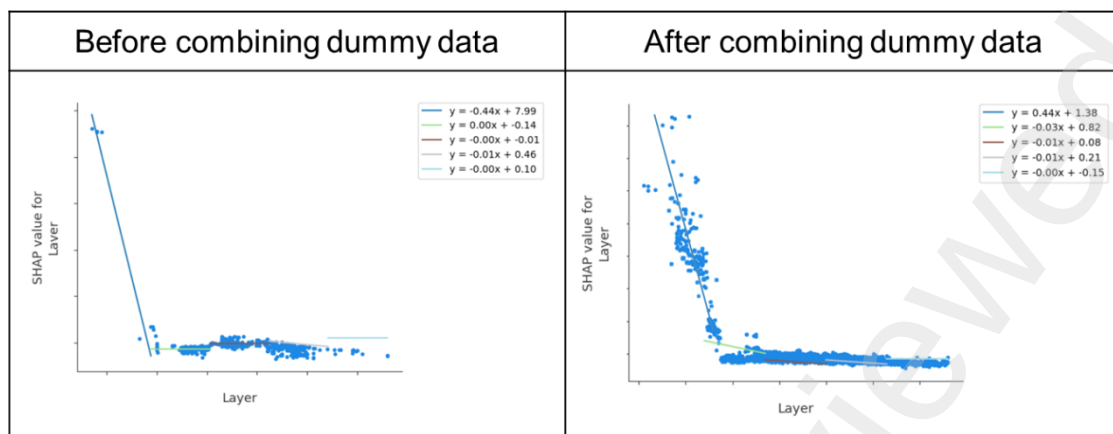


Figure 12 The relation plots between values of feature “Layer” and its SHAP values before and after combining dummy data

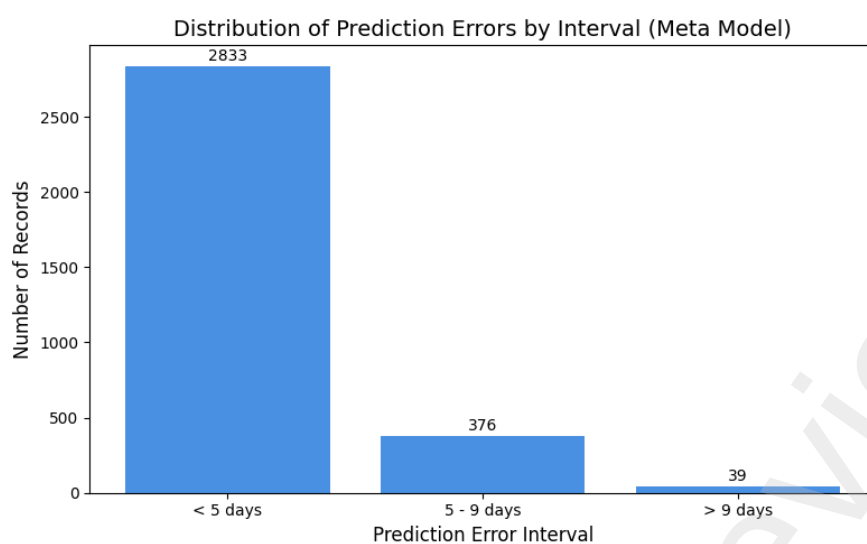


Figure 13 Distribution of prediction errors for meta model

Table 1 Cycle time prediction data

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Table 1 Cycle time prediction data

Categories of information related to cycle time	Included Columns and Descriptions
Status of each machine group	Each production plan will be assigned to specific machine, which belongs to different machine groups and each group has its WIP, MOVE and utilization information.
Categories of products	The product types of columns will be formed by one-hot encoding, which marks the products of each row of data.
Information related to products	The characteristics of production plan about products include quantities of input wafers, priorities of lots...

Table 2 Feature selection for basic models

Model	Feature selection methods	Number of remaining features
XGBoost	Correlation analysis and greedy method based on feature importance	793
Random Forest	Correlation analysis and greedy method based on feature importance	81
SVR	Correlation analysis and greedy method based on feature importance	94
DNN	Principal component analysis (PCA)	331

Table 3 Setting for Hyperparameter tuning of each model

Model	Hyperparameter	Range
XGBoost	n_estimators	[1, 1000]
	learning_rate	[0.001, 0.1]
	max_depth	[1, 10]
	subsample	[0.05, 1.0]
	colsample_bytree	[0.05, 1.0]
	min_child_weight	[1, 20]
Random Forest	n_estimators	[100, 1000]
	max_depth	[5, 30]
	min_sample_split	[2, 20]
	min_sample_leaf	[1, 20]
	max_features	[None, 'sqrt', 'log2']
SVR	kernal	['rbf', 'linear']

Table 4 Comparison of models for validation data in CT prediction

Status	Metrics	XGBoost	RF	SVR	DNN	Meta model	Empirical Methods
Before hyperparameter tuning	MAE (days)	6.2	5.8	12.3	7.1	5.7	7.5
	MAPE (%)	13.5	11.2	23.7	12.4	10.6	11.2
	Improve rate of MAE (%)	17.3	22.6	-64	5.33	24.0	-
After hyperparameter tuning	MAE (days)	5.8	5.1	8.4	5.6	*4.9	7.5
	MAPE (%)	10.3	8.9	18.7	10.3	*8.6	11.2
	Improve rate of MAE (%)	22.6	32.0	-12.0	25.3	*34.6	-

Table 5 Comparison of models for testing data in CT prediction

Metrics	XGBoost	RF	SVR	DNN	Meta model	Empirical Methods
MAE (days)	6.42	7.61	14.21	13.07	7.41	7.5
MAPE (%)	12.9	14.1	25.3	24.2	13.8	11.2
Improve rate of MAE (%)	14.4	-0.014	-89.4	-74.2	0.012	-

Table 6 Comparison of models' performance for testing data before and after combining dummy data

Status	Metrics	XGBoost	RF	SVR	DNN	Meta model	Empirical Methods
Before combining dummy data	MAE (days)	6.42	7.61	14.21	13.07	7.41	7.5
	MAPE (%)	12.9	14.1	25.3	24.2	13.8	11.2
	Improve rate of MAE (%)	14.4	-0.014	-89.4	-74.2	0.012	-
After combining dummy data	MAE (days)	6.13	6.72	9.71	7.19	*6.02	7.5
	MAPE (%)	11.7	12.9	16.4	12.1	*10.6	11.2
	Improve rate of MAE (%)	18.2	10.4	-29.4	4.1	*19.7	-